

# Thabang L. Mashinini-Sekgoto

LEAD DATA SCIENTIST · AI & ANALYTICS ENGINEERING LEADER

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## Summary

Data Science and AI leader with 10+ years of experience delivering enterprise-scale analytics, machine learning, and AI solutions across banking, telecommunications, research, and higher education. Experienced in leading cross-functional teams, operationalising production-grade ML systems, and translating complex technical capabilities into measurable business impact.

Led large-scale AI and analytics initiatives at ABSA, Vodacom, and IBM Research spanning real-time analytics, telematics, optimisation, geospatial intelligence, MLOps, workflow automation, and distributed data platforms. Strong background in Python, SQL, Databricks, cloud platforms, and AI operationalisation with a focus on scalable systems, business enablement, governance, and strategic decision support.

Known for combining deep technical expertise with strong business acumen, stakeholder communication, rapid prototyping, and practical AI implementation in highly regulated enterprise environments.

## Education

### University of the Witwatersrand

Johannesburg, South Africa

PHD IN COMPUTER SCIENCE (IN PROGRESS)

2024 – Present

- Research focus on distributed AI systems, probabilistic modelling, and scalable machine learning.
- Focused on self-supervised learning, AI operationalisation, and distributed computing systems.

### University of the Witwatersrand

Johannesburg, South Africa

MSc IN COMPUTER SCIENCE (DISTINCTION)

2018 – 2019

- Thesis: Learning Level Set Method by Echo State Network for Image Segmentation.
- Developed recurrent neural network approaches for computationally efficient image segmentation.

### University of the Witwatersrand

Johannesburg, South Africa

BSc HONOURS IN COMPUTER SCIENCE

2017

- Project: Wildfire Estimation Using Kernel Density Estimators.

### University of the Witwatersrand

Johannesburg, South Africa

BSc IN COMPUTATIONAL & APPLIED MATHEMATICS AND ASTRONOMY

2014 – 2016

- Focused on mathematical modelling, astrophysics, simulation, and numerical methods.

## Experience

### ABSA Insurance

Johannesburg, South Africa

LEAD DATA SCIENTIST (ACTING HEAD OF DATA SCIENCE)

Mar. 2024 – Present

- Leading enterprise AI, analytics, and data science initiatives within ABSA Insurance.
- Built and operationalised scalable machine learning pipelines on Databricks reducing processing latency by 80–90%.
- Led modernisation of telematics and rewards analytics ecosystems supporting insurance and vehicle finance products.
- Developed geospatial flood-risk analytics models improving underwriting visibility and operational risk assessment.
- Introduced AI governance, workflow standards, monitoring, and reusable analytics frameworks across teams.
- Partnered with engineering, architecture, and business stakeholders to align AI initiatives with operational and strategic objectives.
- Led cross-functional collaboration between Data Science, Engineering, Architecture, and Operations teams.
- Mentored junior Data Scientists and contributed to AI capability development within the organisation.

### Vodacom

Johannesburg, South Africa

SENIOR DATA SCIENTIST

Nov. 2021 – Mar. 2024

- Developed real-time analytics and optimisation systems supporting national telecommunications infrastructure.
- Built streaming AI pipelines using Apache Flink, Kafka, PySpark, and Kubernetes.
- Led development of the Smart Generator Optimisation platform contributing to approximately R1 billion in annual operational savings.
- Designed optimisation frameworks integrating IoT telemetry, real-time alarms, and operational intelligence.
- Mentored Data Scientists and contributed to scalable AI and analytics engineering standards.
- Received the Vodacom Star Award for innovation and engineering excellence.

## IBM Research

Johannesburg, South Africa

MACHINE LEARNING RESEARCH SCIENTIST

Apr. 2020 – Nov. 2021

- Developed machine learning and geospatial analytics solutions for environmental intelligence and climate-risk applications.
- Built predictive analytics systems using TensorFlow and distributed data platforms.
- Developed COVID-19 analytics dashboards supporting public-sector decision-making.
- Collaborated with international research and engineering teams across large-scale AI initiatives.
- Integrated machine learning workflows into the IBM PAIRS Geospatial Platform.
- Co-authored research publications and contributed to global AI research initiatives.

## Business Intelligence Services – University of the Witwatersrand

Johannesburg, South Africa

DATA SCIENTIST

Jun. 2018 – Apr. 2020

- Developed recommendation and analytics systems supporting institutional planning and student success initiatives.
- Built reporting and analytics workflows using Python, SQL, and Power BI.
- Developed clustering-based recommendation systems generating over R2 million annually in government subsidy impact.
- Facilitated analytics and machine learning workshops for technical and non-technical stakeholders.

## Council for Scientific and Industrial Research (CSIR)

Pretoria, South Africa

DATA SCIENTIST AND SOFTWARE ENGINEER

Nov. 2017 – Jan. 2018

- Developed predictive analytics and operational intelligence systems supporting municipalities and public-sector decision-making initiatives.
- Built Django-based decision-support systems enabling operational visibility and real-time access to analytics for public-sector stakeholders.
- Applied machine learning and data engineering techniques to identify operational bottlenecks and improve service delivery planning processes.
- Collaborated within multidisciplinary teams combining software engineering, analytics, and public-sector innovation initiatives.
- Recognised for innovation in predictive modelling and enterprise solutions by  [Mail & Guardian](#),  [CSIR](#), and  [Department of Science and Technology \(DST\)](#).

## Skills

|                                   |                                                                                                 |
|-----------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Leadership</b>                 | AI Strategy, Technical Leadership, Stakeholder Management, Team Mentorship, Agile Delivery      |
| <b>Programming</b>                | Python, SQL, R, JavaScript, C/C++, Shell Scripting                                              |
| <b>AI &amp; Machine Learning</b>  | Machine Learning, Predictive Analytics, NLP, Computer Vision, Prompt Engineering, AI Automation |
| <b>Data Platforms</b>             | Databricks, Apache Spark, Kafka, Flink, Hadoop, ETL/ELT Pipelines                               |
| <b>Analytics</b>                  | Power BI, Dashboarding, Data Visualisation, Business Intelligence, Statistical Analysis         |
| <b>MLOps &amp; Engineering</b>    | MLflow, Docker, Kubernetes, CI/CD, GitHub Actions, Real-time ML Systems                         |
| <b>Workflow Automation</b>        | n8n, AI-assisted scripting, Rapid Prototyping, Process Automation                               |
| <b>Cloud &amp; Infrastructure</b> | AWS, Kubernetes, Linux, Distributed Systems, Cloud-native AI Platforms                          |

## Publications

### Long-Range Seasonal Forecasting of 2m-Temperature with Machine Learning

NEURIPS 2020

2020

- Published research focused on probabilistic forecasting and machine learning methods for climate intelligence systems.
-  [Publication Link](#)

### ML-based Probabilistic Prediction of 2m Temperature and Total Precipitation

IBM RESEARCH

2022

- Research focused on machine learning approaches for probabilistic climate forecasting.
-  [Publication Link](#)

### Mine Worker Threshold Shift Estimation via Optimization Algorithms for Deep Recurrent Neural Networks

IFAC MMM 2019

2019

- Research focused on optimisation and recurrent neural networks in mining applications.

## Selected Projects

## AI Knowledge Platform & Virtual AI Mentor

AI EDUCATOR

2024 – Present

- Built and lead a public AI education and technical leadership platform focused on practical AI adoption, career mentorship, and knowledge accessibility.
- Host AI education and public knowledge-sharing sessions with Prof. Mamokgethi Phakeng focused on real-world applications of AI, automation, machine learning, and modern software engineering.
- Developed technical blogs and educational content simplifying complex AI, distributed systems, MLOps, and machine learning engineering concepts for students, professionals, and business audiences.
- Architected and deployed an interactive AI-powered “Thabang AI” virtual assistant trained on personal technical work, blogs, research, and career experiences to guide aspiring engineers and data scientists.
- Platform demonstrates practical applications of LLMs, retrieval systems, AI-assisted mentorship, and human-centred AI interfaces while positioning AI as an accessible tool for education and empowerment.
-  [Personal AI Platform](#)

## Ubunye AI Ecosystems

FOUNDER & ARCHITECT

2024 – Present

- Designed and lead development of an enterprise-focused open-source AI and ETL ecosystem addressing fragmented ML operationalisation and data engineering workflows within organisations.
- Architected a modular distributed framework supporting Spark, Dask, and Pandas backends enabling scalable ML pipelines, reusable ETL workflows, and production-grade analytics systems.
- Introduced standardised engineering patterns for AI governance, reproducibility, model operationalisation, and scalable feature engineering.
- Framework adopted internally to improve engineering maturity, delivery speed, maintainability, and operational consistency across enterprise AI initiatives.
- Project reflects leadership in AI platform engineering, distributed systems architecture, and scalable ML infrastructure design.
-  [Website Documentation](#)

## Tfilterspy

CREATOR & PRINCIPAL DEVELOPER

2023 – Present

- Developed a scalable Bayesian filtering and probabilistic state-estimation library designed for noisy real-world telemetry, IoT, and time-series environments.
- Implemented distributed Kalman and Particle filtering architectures using Dask to support large-scale streaming and sensor-data processing.
- Built the library to address operational reliability challenges experienced in production systems across insurance, telecommunications, and real-time analytics environments.
- Combined mathematical modelling, distributed computing, and production engineering principles into an open-source framework designed for practical enterprise AI applications.
- Demonstrates deep technical expertise in probabilistic AI systems, numerical computing, distributed systems, and scalable software engineering.
-  [Website Documentation](#)

## Smart Generator Optimisation Platform

SENIOR DATA SCIENTIST

2021 – 2024

- Led development of a national-scale optimisation and operational intelligence platform supporting Vodacom’s telecommunications infrastructure during severe load-shedding periods.
- Designed distributed real-time analytics pipelines integrating IoT telemetry, optimisation algorithms, streaming infrastructure, and operational intelligence systems.
- Built scalable AI and optimisation workflows using Apache Flink, Kafka, PySpark, Kubernetes, and graph-based optimisation techniques.
- Enabled intelligent generator allocation, operational prioritisation, and fuel optimisation across thousands of infrastructure sites nationally.
- Platform contributed to approximately R1 billion in annual operational savings while improving infrastructure resilience and operational visibility.
- Demonstrated technical leadership across distributed systems, optimisation engineering, real-time AI systems, and large-scale enterprise operationalisation.
-  [Vodacom Media Coverage](#)

## Kasi’Lam Digital Platforms

Co-FOUNDER

2022 – Present

- Co-founded digital platform initiatives focused on improving digital accessibility and online presence for township SMEs and local entrepreneurs.
- Designed lightweight, scalable React-based web platforms enabling low-cost digital transformation for underserved communities.
- Focused on combining technology, accessibility, and practical business enablement to reduce barriers to digital adoption.
- Demonstrated leadership in community-focused technology innovation, product thinking, and scalable web engineering.
-  [Project website](#)

## Honors & Awards

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|             |                                                                                                     |                                        |
|-------------|-----------------------------------------------------------------------------------------------------|----------------------------------------|
| 2022        | <b>Vodacom Star Award</b> , Recognised for innovation and engineering excellence                    | <i>Vodacom</i>                         |
| 2019        | <b>MSc with Distinction</b> , Outstanding academic and research performance                         | <i>University of the Witwatersrand</i> |
| 2014 – 2017 | <b>Co-Founder – Wits Astronomy Club</b> , STEM outreach and astronomy public engagement initiatives | <i>Wits University</i>                 |

## References

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|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| Dr.  | <b>Etienne Vos (IBM Research Manager)</b> ,  <a href="mailto:Etienne.vos@ibm.com">Etienne.vos@ibm.com</a>                | <i>IBM Research</i>          |
| Dr.  | <b>Jacobus Du Toit (Machine Learning Lead)</b> ,  <a href="mailto:jacoDutoit@vodacom.co.za">jacoDutoit@vodacom.co.za</a> | <i>Vodacom</i>               |
| Mr.  | <b>Malachy Morrow (Principal Solution Architect)</b> ,  <a href="mailto:mal@malmorrow.com">mal@malmorrow.com</a>         | <i>Altron &amp; Vodacom</i>  |
| Mrs. | <b>Harshila Dulabh (Team Lead)</b> ,  <a href="mailto:Harshila.Dulabh@wits.ac.za">Harshila.Dulabh@wits.ac.za</a>         | <i>Wits University (BIS)</i> |